

Research Methodology

Syllabus

- Unit 1. Introduction to Research Methodology
- Unit 2 :Literature Review and formulation of Research problem
- Unit 3: Research Design
- Unit 4: Hypothesis and Sampling
- Unit 5: Data Collection, Processing and Analysis of Data.
- Unit 6: Interpretation and Report Writing
- Unit 7:Publication ethics and Open access Publishing

Chapter 1

Introduction to Research Methodology

Contents

- 1.1. Meaning of Research 1.2 Objectives of Research 1.3 Motivation in Research 1.4 Types of Research 1.5 Research Approaches 1.6 Significance of Research 1.7 Researcher and Characteristics of Researcher 1.8 Research Ethics and Integrity 1.9 Plagiarism and types of plagiarism 1.10 Introduction to Plagiarism check tools 1.11 Research Methods versus Methodology 1.12 Research and Scientific Method 1.13 Importance of Knowing How Research is Done 1.14 Criteria of Good Research

EXAMPLE OF RESEARCH AND THE SCIENTIFIC METHOD



Why is Stu the pigeon
pecking at the glass,
while Bill the pigeon
is not.

 Study.com®

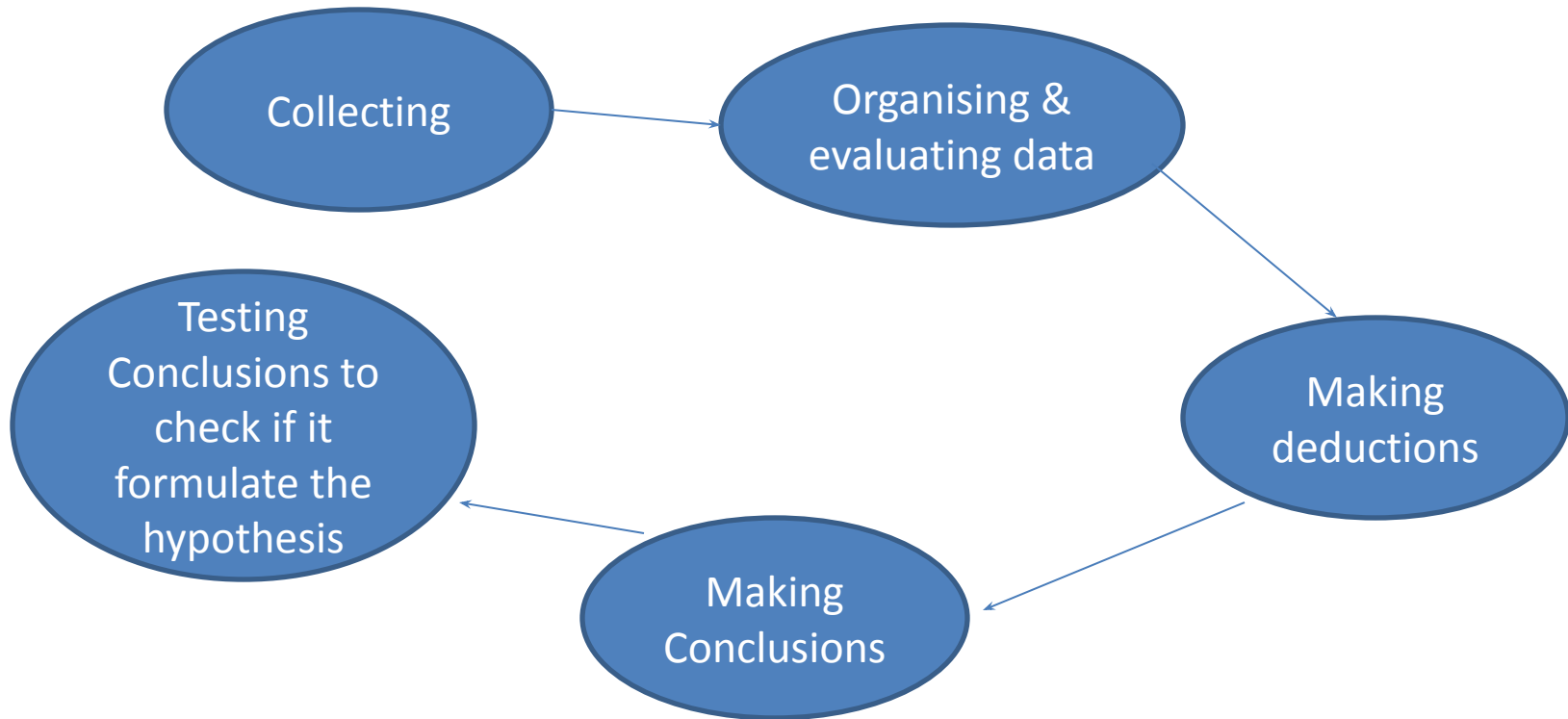
What is Research?

- Search for knowledge.
- Search for knowledge in a systematic manner
- Search for knowledge in a scientific and systematic manner.
- An art of scientific investigation
- A movement from the Known to the Unknown
- A voyage of Discovery

- The Advanced Learner's Dictionary of Current English □ “a careful investigation or inquiry specially through search for new facts in any branch of knowledge.”
- Redman and Mory □ A “systematized effort to gain new knowledge.”

- Research is an academic activity and as such the term should be used in a technical sense.
- According to Clifford Woody
 - Research comprises defining and redefining problems, formulating hypothesis or suggested solutions;
 - collecting, organising and evaluating data; making deductions and reaching conclusions; followed by carefully testing the conclusions to determine whether they fit the formulating hypothesis

Research Definition



- Research refers to the systematic method consisting of
 - enunciating the problem
 - formulating a hypothesis
 - collecting the facts or data
 - Analysing the facts and reaching certain conclusions either in the form of solutions(s) towards the concerned problem or in certain generalisations for some theoretical formulation

Objectives of Research

- The objectives of research can be broadly grouped into following
 - 1. To gain familiarity with a phenomenon or to achieve new insights into it (**Exploratory** or formulative research studies)
 - Eg :Exploring how university students use generative AI tools for academic work — this may reveal patterns, motivations, and concerns that can later lead to a structured survey or experiment.
 - 2. To portray accurately the characteristics of a particular individual, situation or a group (**Descriptive** research studies);
 - Eg:"A Study on the Usage Patterns of Mobile Banking Among College Students in India"

- 3. To determine the frequency with which something occurs or with which it is associated with something else (**Diagnostic** research studies)
 - Eg: Diagnosing the Causes of Low Academic Performance Among First-Year Engineering Students"
- 4. To test a hypothesis of a causal relationship between variables (**Hypothesis-testing** research studies)
 - Eg: Does Daily Exercise Improve Concentration Levels Among College Students?

Difference between the 3 objectives

- **Feature**
 - **Exploratory Research** To explore a problem or idea and gain insights
 - **Descriptive Research** To describe characteristics, behaviors, or patterns
 - **Causal Research Purpose** To determine cause-and-effect relationships
- **Research Question**
 - *What is happening? Why?* (Ex)
 - *What, when, where, how much?* (Des)
 - *Does X cause Y?* (Causal)
- **Hypothesis**
 - Not required
 - May or may not be present
 - Clearly defined and testable
-

- **Approach**
 - Qualitative, open-ended, flexible (Ex)
 - Mostly quantitative,(Des)
 - structured Experimental, highly controlled (Ca)
- **Data Collection Methods**
 - Interviews, focus groups, literature reviews, observations(Ex)
 - Surveys, questionnaires, observations(Des)
 - Experiments, controlled trials, A/B testing(Ca)
- **Outcome**
 - Insights, ideas, problem definition(Ex)
 - Data summaries, trends, population characteristics(Des)
 - Evidence of causality or correlation (Ca)
- **Use Case**
 - Early stage of research
 - Mid-stage to describe the target population
 - Final stage to validate a hypothesis

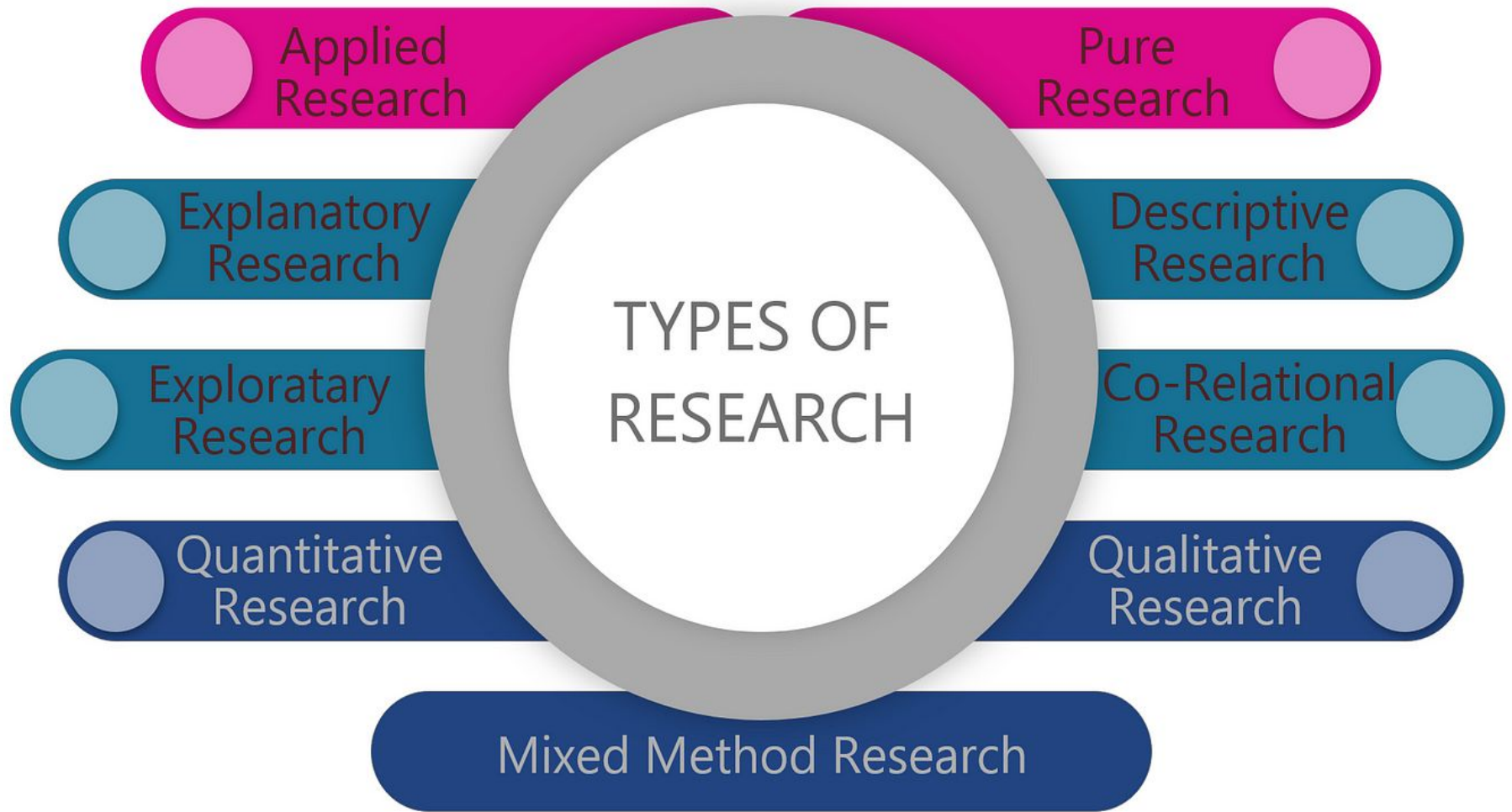
- Descriptive studies focus primarily on what is going on or what exists. (describing something)
 - Eg Public opinion polls , to know what percent of population will vote for a democrat / Republican
- Relational Studies focus on the relationships between two or more variables.
 - Eg comparing proportion of males to females. Wrt voting preference
- Causal studies focus on determining the effect of one or more variables on other variables(the outcome variables)
 - Whether a recent political advertising affected or changed the voter preferences.

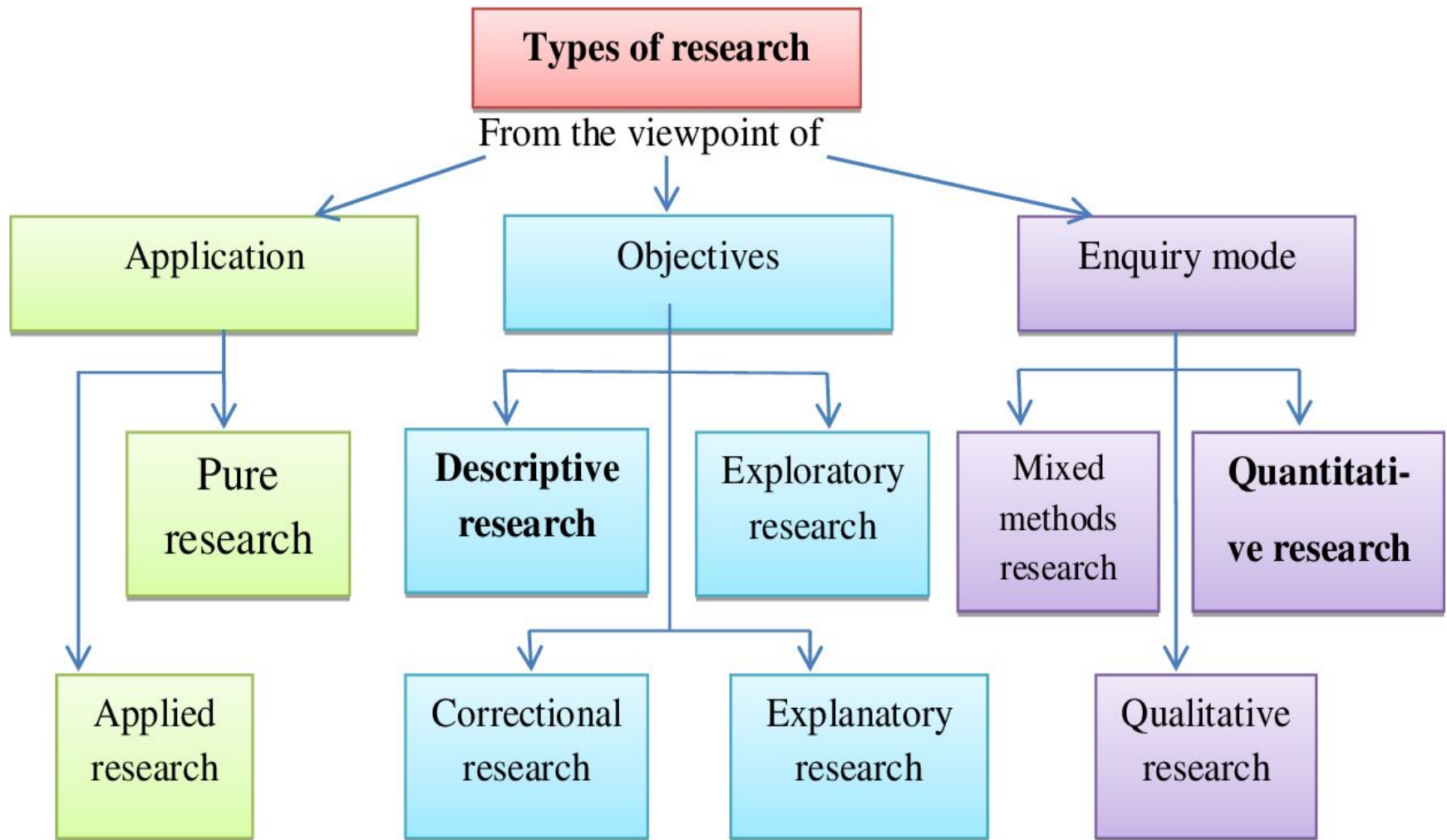
- Above 3 study types may also be combined (and viewed as Cumulative).
 - A relational study implies that we first describe about each variable that we are trying to relate.
 - A causal study assumes that we can describe both the cause and effect variables and that we can show that they are related to each other.
- Causal studies are the most demanding of the three types of studies.

Motivation in Research

- What makes people to undertake research?
 - 1. Desire to get a research degree along with its consequential benefits
 - 2. Desire to face the challenge in solving the unsolved problems,(concern over practical problems initiates research)
 - 3. Desire to get intellectual joy of doing some creative work
 - 4. Desire to be of service to society
 - 5. Desire to get respectability

- Many more factors such as directives of government, employment conditions, curiosity about new things, desire to understand causal relationships, social thinking and awakening, and the like may as well motivate people to perform research operations.





Explanatory □ Causal research; Correlational □ measure the relationship/association between the variables

Types of Research

- 1. Descriptive Vs Analytical
 - *Descriptive research is a methodology that focuses on explaining “what” is happening in a given situation, problem, or phenomenon.*
 - *It only describes and understands a specific topic without explaining the reasons behind whatever is happening.*
 - *It provides information about the characteristics and behaviors but doesn’t give in-depth knowledge of the causes or reasons for those behaviors.*
 - The major purpose of descriptive research is description of the state of affairs as it exists at present.
 - The main characteristic of this method is that the researcher has no control over the variables; he can only report what has happened or what is happening

What is Descriptive Research?



Example



- For example, a travel agency wants to gain information about the travel preferences of its customers.
- They interview a sample of customers and only collect the “what” part of the data, like what are their favorite travel destinations, what accommodations they prefer, and what activities they enjoy.
- The agency does not collect any information about why the customers prefer those destinations, accommodations, and activities.

- **Purpose – Why Perform Descriptive Research?**
 - **Describing Characteristics:** It helps in providing a detailed and objective description of a group, event, or object, giving a clear picture of the topic.
 - **Finding Relationships:** Although it doesn't explain any causes or reasons, it can help make connections or find relationships between different factors.
 - **Identifying Trends:** Researchers can use it to spot patterns or changes over time, helping them understand shifts in a particular area of interest.
 - **Foundation for Analytical Research:** It gathers foundational data before more in-depth research or experiments, helping analytical researchers.
 - **Creating Profiles:** This research helps create profiles of individuals or groups, capturing their characteristics, which is useful in understanding specific populations.
 - .

- **Market Research and Decision Making:** It aids businesses in analyzing consumer behavior, market trends, and customer preferences to make informed decisions.
- **Guiding Policies:** Policymakers and organizations use it to do planning and make decisions based on objective data and evidence

- **Examples of Descriptive Research**

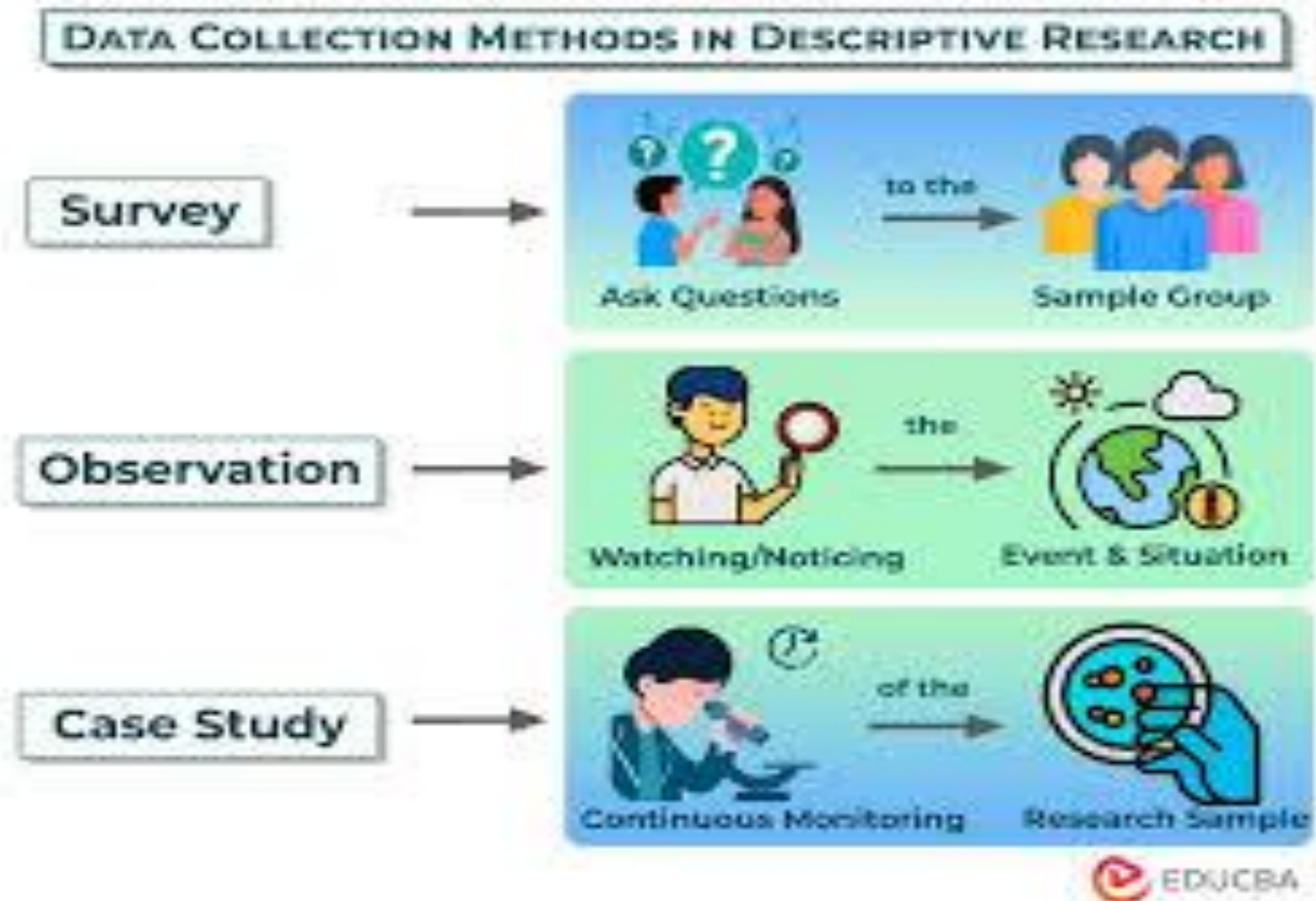
- #1: Census Data Analysis**

- When governments conduct censuses to gather detailed demographic information about their population, they use descriptive research. The data collected includes age, gender, ethnicity, education, occupation, and more. The authorities can analyze this data to describe the population's characteristics, such as the distribution of age groups, gender ratios, or educational attainment in different regions.

- **#2: Workplace Observations**

- Industrial and organizational psychologists may conduct workplace observational studies to describe employee behaviors, job performance, and corporate culture. This descriptive research can provide insights into factors that impact productivity, job satisfaction, and employee well-being.
- **#3: Sleep Habits and Technology Use**
- Sleep researchers may conduct surveys or studies to describe how technology usage before bedtime affects people's sleep quality and duration. This research is particularly relevant in the context of modern technology's impact on sleep patterns.

- **Data Collection Methods of Descriptive Research**



- **How to Conduct Descriptive Research?**
- **Step 1: Choose your Research Topic**
 - Pick a subject you want to study and define what you want to learn about it.
 - Make sure it's something you can realistically investigate and relevant to your field of interest.
- **Step 2: Read Existing Studies**
 - Look at what others have already researched about your topic. It will help you understand the subject better, find gaps in knowledge, and refine your research questions.
- **Step 3: Decide how to Collect Information**
 - Decide how you will gather information. You can use surveys, questionnaires, interviews, observations, case studies, or analyze existing data. Choose the method that suits your topic best.
- **Step 4: Choose Your Sample**
 - Pick a group representing the larger population, so your findings can apply to more people. Also, make sure to use random or stratified sampling to avoid bias.
- **Step 5: Gather and Organize the Data**
 - Collect information following your chosen method. Be careful and accurate in recording the data. You can collect data in person, online, or through other appropriate ways. Finally, arrange your data in an organized manner.

- **Step 6: Draw Conclusions**

- Look at what you found in your data analysis. Explain the key findings and what they mean. Based on your findings, come to conclusions that answer your research questions. Relate your results to what others have found in previous studies.

- **Step 7: Share Your Results**

- Write a detailed report with an introduction, what others have studied before, your methods, findings, and conclusion. Use tables, graphs, and visuals to make it easier to understand.

-

- **A case study example**

- **Scenario:**

Geologist Dr. Richard wants to learn about volcanic activity in the Hawaiian Islands, an area known for its active volcanoes and geological instability. He wants to understand how these volcanoes behave.

- Dr. Richard selects the volcanoes in the Hawaiian Islands as the subject of his study. He aims to find out why these volcanoes erupt and what makes them different from other volcanoes.
 - **Step 2: Finding and Studying Existing Research**
 - Dr. Richard seeks research paper help and reads many books and research papers about the volcanoes in the Hawaiian Islands. He looks at old records and talks to other experts to learn more about these volcanoes.
 - **Step 3: Deciding which Methods to Use**
 - He decides to use the observation method to collect data. He also carries several measuring tools to assist with the data collection.
 - **Step 4: Selecting a Sample**
 - Dr. Richard wants to study different types of volcanoes to understand them better. So, he picks a few different volcanoes to study in the Hawaiian Islands.
 - **Step 5: Collecting the Required Data**
 - Dr. Richard goes to the Hawaiian Islands to study the volcanoes and observes them closely. He also uses tools to measure shaking on the ground caused by volcanic activity.

- **Step 6: Arriving at Conclusions**

- After studying the data, Dr. Richard records all the information collected about the volcanoes in the Hawaiian Islands and explains their behavior.

- **Step 7: Sharing the Research Findings**

- Dr. Richard writes a report with all the information he finds. He shares this report with other scientists, government agencies, and environmentalists.

- **Characteristics of Descriptive Research**
- **Quantitative and qualitative data:**
 - It can use quantitative (numerical) and qualitative (non-numerical) data. Quantitative methods focus on surveys, questionnaires, and statistical analysis, while qualitative methods involve interviews, focus groups, and content analysis.
- **Cross-sectional design:**
 - This research typically has a cross-sectional design, where data is collected at a single point in time. It allows researchers to get a snapshot of the population or phenomenon of interest.
- **Sample selection:**
 - To gather accurate results, researchers use different sampling techniques to choose a sample representing the whole group they want to study. The size and representativeness of the sample are essential to draw conclusions that apply to the entire population.
- **Non-experimental:**
 - Unlike experimental research, which tests cause-and-effect relationships by changing things, descriptive research simply observes and describes what is already happening without changing anything.

- **Data analysis:**

- It involves data analysis techniques such as frequencies, percentages, averages, and measures of central tendency (e.g., mean, median, mode). It helps summarize and describe the data collected.

- **Report findings:**

- The researcher presents the research results descriptively, often through tables, charts, graphs, and narrative descriptions. The help is providing a clear and comprehensive account of the observed data.

- **Preliminary exploration:**

- This research is like a starting point for more studies. It helps researchers find patterns and relationships that might need more research in the future. It's like a clue that can lead to new discoveries.

- **Advantages**

- Provides a detailed and comprehensive understanding of the subject
- Suitable for exploring new topics
- Doesn't invade people's privacy
- Findings can often apply to a larger group
- Time and cost-effectiveness compared to other research types.

- **Disadvantages**

- Can't tell us why things happen, can only tell what happens
- Doesn't explain the reasons behind the findings
- Biases can influence results
- Lack of control over variables
- Findings might not be accurate for a bigger group if the sample is too small or not representative.

Analytical Research

- This particular kind of research calls for using critical thinking abilities and assessing data and information pertinent to the project at hand.
- Determines the causal connections between two or more variables.

Eg : *Analyzing the Effectiveness of Online Learning Platforms in Improving Student Performance in Computer Science Courses“*

Analyzing vaccination data to study patterns in disease reduction

- Importance of analytical research
 - *The goal of analytical research is to develop new ideas that are more believable by combining numerous minute details.*
 - The analytical investigation is what explains why a claim should be trusted. Finding out why something occurs is complex.
 - This kind of information aids in proving the validity of a theory or supporting a hypothesis. It assists in recognizing a claim and determining whether it is true.

- Analytical Research
 - **Objective** : Analyze & interpret data to understand relationships or causality
 - **Focus** : “WHY” and “HOW” questions
 - **Data Analysis**: Statistical research, hypothesis testing, qualitative research.
 - **Goal** : Gain insights, make inferences, provide explanations, predictions
 - **Causal relationships** : Examining underlying factors, causes, or effects

- **Descriptive vs analytical research**

- Objective□

- Descriptive □ Describe and document characteristics or phenomena.
 - Analytical □ Analyze and interpret data to understand relationships or causality.

- Focus□

- Descriptive□ Focus“What” questions
 - Analytical □ “Why” and “How” questions

- Data analysis

- Descriptive □ Summarizing information
 - Analytical □ Statistical research, hypothesis testing, qualitative research

- Goal
 - Descriptive ☐ Provide an accurate and comprehensive description
 - Analytical ☐ Gain insights, make inferences, provide explanations or predictions
- Causal Relationships
 - Descriptive ☐ Not the primary focus
 - Analytical ☐ Examining underlying factors, causes, or effects
- Examples
 - Descriptive ☐ Surveys, observations, case-control study, content analysis
 - Analytical ☐ Examples ☐ Experiments, statistical research, qualitative analysis

Applied research



- Applied research, is designed specifically to solve a problem in the world.
- The applied science researcher takes a need in society and attempts to fix that need.
- Applied research can include things such as reducing the use of nonrenewable resources by finding other alternatives to achieve the same output, cure a disease, or invent a new form of technology to make a process more efficient.
- Because of the potential to fix a problem, applied research tends to have much more commercial value than basic research.
- The potential for monetary gain makes the field of applied research more in-demand by governments and industries.

- Applied Vs Fundamental (Basic research)
 - ● Basic research expands topics we already know; Applied research is used to solve real-life issues
 - ● Basic research is theoretical; Applied research is often tactile and always practical
 - ● Basic research can be used to generate theories that are universally applicable; Applied research is only used to fix the problem at hand
 - ● Basic research intends to develop more knowledge in the field of science and aid in predictions; Applied research focuses on developing technology, efficiency, and other improvements
 - ● The goal of basic research is to add knowledge to what we already know with no intent to use it beyond more information; the goal of applied research is to find a solution that we don't know to a problem that exists

- A few examples of applied Research
 - Analyzing what type of prompts will inspire people to volunteer their time to charities
 - Investigating if background music in a work environment can contribute to greater productivity
 - Investigating which treatment approach is the most effective for reducing anxiety
 - Researching which strategies work best to motivate workers
 - Studying different keyboard designs to determine which is the most efficient and ergonomic

- A hospital may conduct applied research to figure out how to best prepare patients for certain types of surgical procedures.
- A business may hire an applied psychologist to assess how to design a workplace console to maximize efficiency and productivity while minimizing worker fatigue and error.
- An organization may hire an applied researcher to determine how to select employees that are best suited for certain positions within the company.

- Another example
- A social psychologist may perform basic research on how different factors may contribute to violence in general.
- But if a social psychologist were conducting applied research,
 - they may be tackling the question of what specific programs can be implemented to reduce violence in school settings.

- Quantitative vs. Qualitative:
 - Quantitative research is based on the measurement of quantity or amount. It is applicable to phenomena that can be expressed in terms of quantity.
 - Qualitative research, is concerned with qualitative phenomenon, i.e., phenomena relating to or involving quality or kind.
 - Eg :. Investigating the reasons for human behaviour (In turn to support Motivation Research',) an important type of qualitative research

- . Qualitative research is specially important in the behavioural sciences where the aim is to discover the underlying motives of human behaviour.
- Through such research we can analyse the various factors which motivate people to behave in a particular manner or which make people like or dislike a particular thing.

- Conceptual vs. Empirical:
 - These are two ways of doing logical research
 - Conceptual research doesn't include any tests, and empirical investigation does.
 - Conceptual research relates to some abstract idea(s) or theory. (New thoughts and theory)
 - It is generally used by philosophers and thinkers to develop new concepts or to reinterpret existing ones
 - does not involve conducting any practical experiments.
 - .

Conceptual Research

- Conceptual research is defined as a methodology wherein research is conducted by observing and analyzing already present information on a given topic.
- Conceptual research doesn't involve conducting any practical experiments.
- It is related to abstract concepts or ideas.
- Philosophers have long used conceptual research to develop new theories or interpret existing theories in a different light.

- Conceptual research framework constitutes of a researcher's combination of previous research and associated work and explains the occurring phenomenon.
- It systematically explains the actions needed in the course of the research study based on the knowledge obtained from other ongoing research and other researchers' points of view on the subject matter.

- Based on observing and analyzing already existing concepts and theories.
- The researcher can observe their surroundings and develop brand-new theories, or they can build on existing ones.
- Eg : 1. Newton theory of gravity.
 - Nicolaus Copernicus findings that the planets revolve around the sun and not vice versa.
- Conceptual research is widely used in the study of philosophy to develop new ideas.
- Philosophical research studies are cases of conceptual research, while empirical research incorporates both quantitative and subjective studies

- Eg:

- A researcher wants to explore the key factors that influence consumer behavior in the online shopping environment.
- Once the researcher decides on the subject, they can begin by reviewing the existing literature on consumer behavior and examining different theories and models of consumer behavior.
- Then, they can identify common themes or factors that have emerged.
- By understanding this phenomenon, the researcher can develop a conceptual framework.

- A group of researchers wants to see if there is any correlation between chemically dyeing your hair and the risk of cancer in women.
- They can start collecting data on women that had cancer and usage of hair dye. They can collect research papers on this particular subject.
- And they can create a conceptual framework with the information they collected and analyzed.

- Empirical Research (experimental research)
 - Empirical means gained through experience.
 - Includes investigation based on perception, tests, and unquestionable evidence.
 - Research based on observations, experiments and verifiable evidence.
 - It is data-based research, coming up with conclusions which are capable of being verified by observation or experiment.
 - The researcher must first provide himself with a working hypothesis or guess as to the probable results.
 - Next is to get enough facts (data) to prove or disprove his hypothesis.

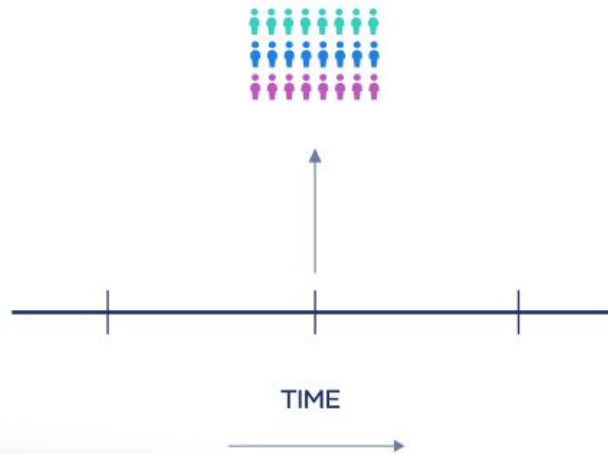
- Next is to set up experimental designs which he thinks will manipulate the persons or the materials concerned so as to bring forth the desired information.
- Empirical research is thus characterized by the experimenter's control over the variables under study and his deliberate manipulation of one of them to study its effects.
- Empirical research is appropriate when proof is sought that certain variables affect other variables in some way.
- Evidence gathered through experiments or empirical studies is today considered to be the most powerful support possible for a given hypothesis

- Eg : An experiment could be conducted where one group of the audience is exposed to happy music and the other is not exposed to music at all.
- A correlational research to understand if high education helps in obtaining better-paying jobs.
 - If outcomes indicate that higher education does allow individuals to have high-salaried jobs, then it follows that people with less education tend to have lower-paying jobs

- There are some other types of research that are combination of the above types:
 - Time based :
 - One-time research (research confined to a single time period)
 - Longitudinal research (research carried over several time periods)
 - In a longitudinal study, researchers repeatedly examine the same individuals to detect any changes that might occur over a period of time.
 - Longitudinal studies are a type of correlational research in which researchers observe and collect data on a number of variables without trying to influence those variables.

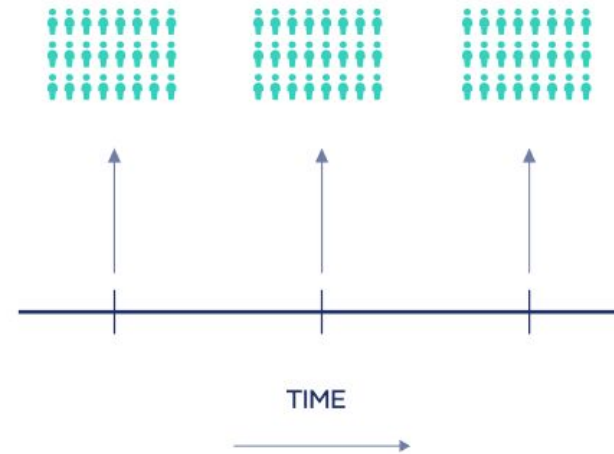
Cross-sectional study

Data collected at one point in time



Longitudinal study

Data collected repeatedly over time



- Example

- To study the relationship between smoking and stomach cancer.
- We first conduct a cross-sectional study to see if there is a link between smoking and stomach cancer, and we discover that a link exists in men but not in women.
- We then decide to design a longitudinal study to further examine this relationship in men.
- Without the cross-sectional study first, we would not have known to focus on men in particular.

- Data from the 1970 British Cohort Study, which has followed the lives of 17,000 Brits since their births in a single week in 1970,
- **Collecting your own data for longitudinal study**
 - Data collection depends on the type of longitudinal study you choose to perform.
 - It can be to conduct a retrospective or a prospective study.
 - In a **retrospective study**, you collect data on events that have already happened.
 - In a **prospective study**, you choose a group of subjects and follow them over time, collecting data in real time.
 - Retrospective studies are generally less expensive and take less time than prospective studies, but are more prone to measurement error.

- Considering our previous example,
 - In a retrospective study, you might look at past medical records of patients to see whether those who developed this cancer had previously smoked.
 - In a prospective study, you might follow a group of both smokers and non-smokers over time to see if they develop cancer later on.

- Research can be field-setting research or laboratory research or simulation research, depending upon the environment in which it is to be carried out.
- Research can as well be understood as clinical or diagnostic research.
 - Such research follow case-study methods or in-depth approaches to reach the basic causal relations.
 - Such studies usually go deep into the causes of things or events that interest us, using very small samples and very deep probing data gathering devices

- Research can be Historical research, which utilizes historical sources like documents, remains, etc. to study events or ideas of the past, including the philosophy of persons and groups at any remote point of time.
- Research can also be classified as conclusion-oriented and decision-oriented.
 - While doing conclusion-oriented research, a researcher is free to pick up a problem, redesign the enquiry as he proceeds and is prepared to conceptualize as he wishes.
 - Eg : Though there has been much debate on the subject, it is clear that democratic leadership is the best form of management for the modern workplace.

- Decision-oriented research is always for the need of a decision maker and the researcher in this case is not free to embark upon research according to his own inclination.
 - Operations research is an example of decision oriented research since it is a scientific method of providing executive departments with a quantitative basis for decisions regarding operations under their control

Research Approaches

- The research approach is a plan and procedure that consists of the steps of broad assumptions to detailed methods of data collection, analysis, and interpretation.
- It is, therefore, based on the nature of the research problem being addressed.
- The research approach is essentially divided into two categories:
 - the approach of data collection and
 - the approach of data analysis or reasoning.



- Based on the data collection, there are two basic approaches to research:
 - Quantitative Approach □ generation of data in Quantitative form, and subject to rigorous quantitative analysis.
 - Qualitative Approach □ subjective assessment of attitudes, opinions and behaviour.

- Quantitative can be sub-classified into
 - Inferential, experimental, and simulation approaches.
 - The purpose of **inferential approach** to research is to form a data base from which to infer characteristics or relationships of population.
 - Eg. A survey research where a sample of population is studied (questioned or observed) to determine its characteristics, and it is then inferred that the population has the same characteristics

- **Experimental approach** is characterized by much greater control over the research environment and in this case some variables are manipulated to observe their effect on other variables.
- **Simulation approach** involves the construction of an artificial environment within which relevant information and data can be generated.

- Qualitative approach
 - Research in such a situation is a function of researcher's insights and impressions.
 - Involves collecting and analyzing non-numerical data (e.g., text, video, or audio) to understand concepts, opinions, or experiences.
 - It can be used to gather in-depth insights into a problem or generate new ideas for research.
 - Opposite of [quantitative research](#), which involves collecting and analyzing numerical data for statistical analysis.
- Qualitative research is commonly used in the humanities and social sciences, in subjects such as anthropology, sociology, education, health sciences, history,

- Qualitative research question examples
 - How does social media shape body image in teenagers?
 - How do children and adults interpret healthy eating in the UK?
 - What factors influence employee retention in a large organization?
 - How is anxiety experienced around the world?
 - How can teachers integrate social issues into science curriculums?

- Qualitative research approaches:
 - Grounded theory □ Researchers collect rich data on a topic of interest and develop theories inductively.
 - Ethnography □ Researchers immerse themselves in groups or organizations to understand their cultures.
 - Action research □ Researchers and participants collaboratively link theory to practice to drive social change.
 - Phenomenological research □ Researchers investigate a phenomenon or event by describing and interpreting participants' lived experiences.
 - Narrative research □ Researchers examine how stories are told to understand how participants perceive and make sense of their experiences.

Deductive and Inductive Approaches

Deductive Approach

A social phenomenon is observed.



A theory is developed to explain why it occurred.



The theory is tested through research and the theory is either accepted, rejected, or revised.

Inductive Approach

A social phenomenon is observed.



Data is collected on the possible reasons why it occurs and trends in the data are examined.



A theory is developed from this data to explain the social phenomenon.

- Inductive approach □ begins with a set of experimental observations, seeking patterns in those observations, and then theorizing those patterns.
- Deductive approach □ begins with a theory, developing hypotheses from that theory, and then collecting and analyzing data to test those hypotheses.

- **Deductive** reasoning depends on facts and evidence; **inductive** reasoning looks at patterns.
- **Deductive** reasoning provides solid, repeatable conclusions. **Inductive** reasoning makes general, most probable conclusions about evidence that has been observed.
- **Inductive** reasoning may not always have strong conclusions on the validity of its hypothesis. **Deductive** reasoning will always have strong conclusions as to whether the premise is valid or invalid.
- **Deduction** moves from idea to observation, while **induction** moves from observation to idea.
- **Deduction** moves from more general to more specific, while **induction** moves from more specific to more general.

- **Deductive** arguments have unassailable conclusions assuming all the premises are true, but **inductive** arguments simply have some measure of probability that the argument is true—based on the strength of the argument and the evidence to support it.
- The main difference between **inductive** and **deductive** approaches to research is that **inductive** approach is concerned with the generation of new theory emerging from the data, where as **deductive** approach starts with a statement or hypothesis and then tests to see if it's true through observation.

Significance of Research

- Significance ☐
 - “All progress is born of inquiry”
 - “Doubt is often better than overconfidence, for it leads to inquiry, and inquiry leads to invention”
- Research inculcates scientific and inductive thinking and it promotes the development of logical habits of thinking and organisation.

- Research provides the basis for nearly all government policies in our economic system.
 - Decision-making may not be a part of research, but research certainly facilitates the decisions of the policy maker.
- Another area in government, where research is necessary, is collecting information on the economic and social structure of the nation.
 - Such information indicates what is happening in the economy and what changes are taking place

- Research as a tool to economic policy has three distinct phases of operation :
 - (i) investigation of economic structure through continual compilation of facts
 - (ii) diagnosis of events that are taking place and the analysis of the forces underlying them
 - (iii) the prognosis, i.e., the prediction of future developments
- .

- Research has its special significance in solving various operational and planning problems of business and industry.
 - Operations research and market research, along with motivational research, are considered crucial and their results assist, in more than one way, in taking business decisions.
 - Market research is the investigation of the structure and development of a market for the purpose of formulating efficient policies for purchasing, production and sales.

- Research is equally important for social scientists in studying social relationships and in seeking answers to various social problems.
 - It provides the intellectual satisfaction of knowing a few things just for the sake of knowledge and also has practical utility for the social scientist to know for the sake of being able to do something better or in a more efficient manner.
 - Research in social sciences is concerned both with knowledge for its own sake and with knowledge for what it can contribute to practical concerns

- In addition to what has been stated above, the significance of research can also be understood keeping in view the following points:
 - (a) To those students who are to write a master's or Ph.D. thesis, research may mean a careerism or a way to attain a high position in the social structure;
 - (b) To professionals in research methodology, research may mean a source of livelihood;
 - (c) To philosophers and thinkers, research may mean the outlet for new ideas and insights;
 - (d) To literary men and women, research may mean the development of new styles and creative work;
 - (e) To analysts and intellectuals, research may mean the generalisations of new theories.
- Thus, research is the fountain of knowledge for the sake of knowledge and an important source of providing guidelines for solving different business, governmental and social problems. It is a sort of formal training which enables one to understand the new developments in one's field in a better way.

Characteristics of a Researcher

- **Curiosity:** A curious mind and an ability to look at things from different perspectives is what makes a good researcher better.
- **Critical thinking:** Successful researchers can think critically about the information they gather while reading about new developments in their own and related fields

- **Creativity:** The qualities of a good researcher do not just include curiosity and critical thinking, but also thinking creatively when it comes to problem solving.
- **Objectivity:** Nurturing preconceived notions is detrimental to research. Avoid temptations to make unconclusive statements or introduce personal biases into research, which will impact your research and standing in the long run.

- **Collaborative spirit:** An important characteristic of a good researcher is being able to work well with others.
- **Communication skills:** An added strength of a good researcher is being able to communicate your findings clearly and effectively, which is a key contributor to your success

- **Attention to detail:** One of the key qualities of a good researcher is being meticulous in your work.
- **Time management:** Most successful researchers organize, prioritize, and optimize their time efficiently, allowing them to not only keep up with their responsibilities but also make time for personal tasks.

- **Persistence & flexibility:** Research can be a long, difficult process with several hurdles and changes along the way. One of the key requirements to becoming a good researcher is being able to adapt to new technologies and changing circumstances and persevere despite setbacks and challenges that inevitably arise.
- **Focus on self-care:** Anxiety, stress, and mental health issues are common among academics.

Research ethics and Integrity

- Research Ethics is defined as the ethics of the planning, conduct, and reporting of research.
- Research ethics provides researchers with the guidelines to conduct their research responsibly.
- Provides A code of guidelines on how to conduct scientific research in a morally acceptable way.
- Provides Principles and standards that help researchers to uphold the value and standards of knowledge construction

- Research ethics involves :
 - The application of basic ethical concepts to research activities, such as
 - The design and implementation of research
 - Respect for society and other
 - The use of resources and research output
 - Scientific misconduct
 - Research regulation

- Three Objectives of Research Ethics
 - Protecting human participants
 - Focus on issues concerning biomedical research.
 - Focus on concerns related to vulnerable groups of people (people who are institutionalized / people with developmental disabilities/ people without legal status /homeless people etc)

- Principles of Research Ethics

- Honesty □ Honesty in reporting data, research methods, and procedures, results, and publication status. Never falsify, fabricate or misrepresent data.
- Objectivity □ always aim to avoid biases in experimental design, data analysis and interpretation, personal decisions, expert testimony, peer review and all other aspects of research
- Integrity □ do not act contrary to your promises and agreements.

- Carefulness □ avoid negligence and making careless errors. Keep the records for every research activity.
- Respecting Intellectual property □ always respect copyrights, patents and any other type of intellectual property.
- Confidentiality □ safeguard confidential information like personal information, military or trade secrets, and patient records.

- Dos and Don'ts of Research Ethics
 - **Do** Maintain a good record of all your research activities and report your data as carefully and objectively as possible.
 - **No** Fabrication, manipulation or misrepresentation of data.
 - **Do** Disclose financial or personal interests that may directly/indirectly affect your work.
 - **No** Deceiving research sponsors, colleagues, or ethical committees by having bias in data interpretation, peer review, or personnel decisions.

- **Do** Treat animals with care and respect when studying them in your research and adhere to ethical guidelines.
- **Do not** Use any external research data (published or unpublished) without permission.
- **Do** Respect intellectual property, privacy, and confidentiality and give proper credit for any contributions from other researchers.
- **Do not** Support irresponsible publication practices. Your main goal should be to advance science and share your knowledge within the community

Plagiarism and types of plagiarism

- Plagiarism includes
 - Copying and pasting from a source without attaching the text to quotation marks and giving a citation.
 - Summing up or rephrasing another person's thoughts without giving a reference.
 - Submitting an assignment completed by someone else.
 - Reusing the assignment, you have submitted for the previous course. This is called self-plagiarism.

- Working on an assignment with a friend or classmate that is meant to be finished individually.
- Writing a paper that strings together quote after quote or paraphrase, regardless of whether referred to effectively.

• 7 types of plagiarism

- **Complete plagiarism** : This type of plagiarism occurs when a writer submits someone else's work in their own name.
 - Paying somebody to write a paper for you, then handing that paper in with your name on it, is an act of complete plagiarism.
- **Direct Plagiarism** : Similar to complete plagiarism in that it, it is the overt passing-off of another writer's words as your own. But here its how much of the paper is copied .
 - With complete plagiarism, it's the entire paper. With direct plagiarism, specific sections or paragraphs are included without crediting (or even acknowledging) the author.

- **Paraphrasing plagiarism** : Paraphrasing plagiarism is what happens when a writer reuses another's work and changes a few words or phrases.
- **Self Plagiarism** : Reusing content from one of our own work into another research paper would be an act of self-plagiarism
- **Patchwork Plagiarism** : Also known as mosaic plagiarism, patchwork plagiarism refers to instances where plagiarized work is interwoven with the writer's original work.
 - This kind of plagiarism can be subtle and easy to miss, and it may happen in conjunction with direct plagiarism.

- **Source-based plagiarism** : With this kind of plagiarism, the writer might cite their sources correctly but present the sources in a misleading way.
 - For example, the writer might reference a secondary source in their work but only credit the [primary source from which that secondary source is derived](#).

- **Accidental Plagiarism** : The most common type of plagiarism because it happens when the writer doesn't realize they are plagiarizing another's work. Accidental plagiarism includes the following:
 - Forgetting to [cite your sources](#) in your work
 - Not citing your sources correctly
 - Failing to put quotes around cited material

- **Consequences of plagiarism**
- Plagiarism can have serious consequences. Depending on the nature of the plagiarism and the university or instructor's policy, here are some possible consequences:
 - Academic probation
 - Failure of the assignment
 - Failure of the course
 - Suspension
 - Dismissal from the program or the institution

Introduction to Plagiarism check tools

- Plagiarism tools are the modern tools that we use to detect the portions of duplicate or copied content in everything from research journals to site content.
- They are very quick, exact and can work in numerous languages.
- The AI-powered tools further have progressed multi-layered search mechanisms, via which they can wade through billions of private documents, scientific and medical journals and public databases.
- Plagiarism tools also access those websites which are password protected and that is not easily accessible.

1. Dupli Checker
2. Copy Leaks
3. Plagiarisma
4. Plagiarism Checker
5. Plagium
6. PlagScan
7. PlagTracker
8. Quetext
9. PaperRater
10. Plagiarismhunt

Research Methods versus Methodology

Research Methodology	vs	Research Method
Refers to a process for effectively addressing research problems		Describes the methods that the researcher uses to conduct the research
Investigate the various methods that can be used to conduct experiments, tests, surveys		It involves conducting an experiment, test, survey
Used to find a solution to the research problem		Used to achieve the research objective
Aims to answer this question. How did the researcher complete his study?		Aims to answer this question. What tools did the researcher use to complete his research?
Any research must start with research methodology because it provides the way for choosing the most suitable research methods		Research, whether it be scientific or not, must have research methods

- **Example of Research Methodology**

- A qualitative case study using data from observational field notes and interviews is an example of research methodology.

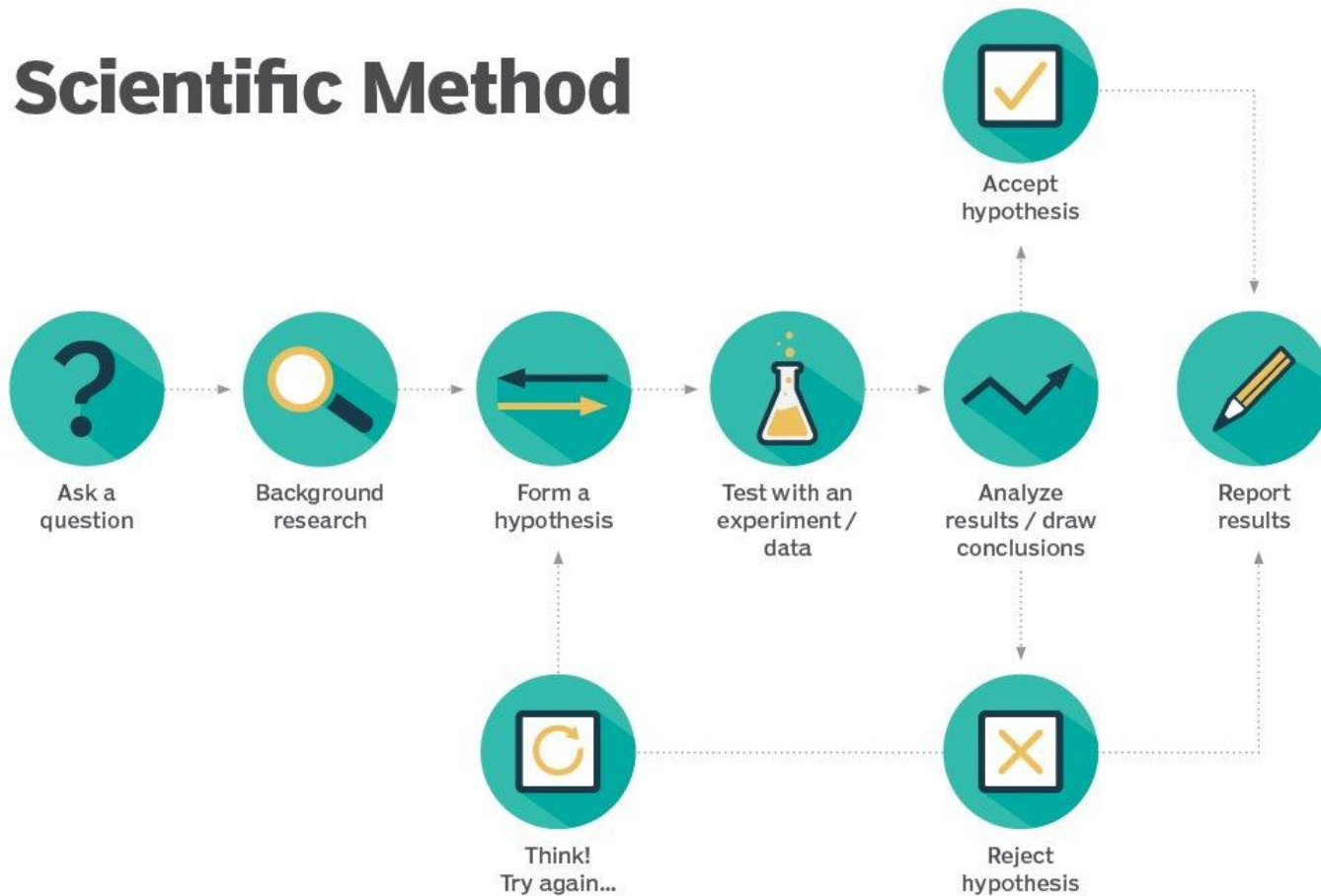
- **Example of Research Method**

- A physical scientist, for instance, might use equipment like an electron microscope or a radio telescope to gather his data.
- A researcher or social scientist, on the other hand, might gather his data using a personal interview, a sample survey with a mail questionnaire, or an opinion poll as a method.

- **Research and Scientific method**

- Research includes any formal gathering of data, information and facts for the advancement of knowledge.
- Research is conducted according to the researcher's intention, their purpose, and the paradigm they're operating from within.
- The scientific method is the process of objectively establishing facts through testing and experimentation.
 - The basic process involves making an observation, forming a hypothesis, making a prediction, conducting an experiment and finally analyzing the results

Scientific Method



Importance of knowing the research process

- The knowledge of methodology provides good training specially to the new research worker and enables him to do better research.
- Helps the researcher to develop disciplined thinking or a “bent of mind” to observe the field objectively.
- Knowledge of how to do research will inculcate the ability to evaluate and use research results with reasonable confidence.

- When one knows how research is done, then one may have the satisfaction of acquiring a new intellectual tool which can become a way of looking at the world and of judging every day experience.

Criteria of good research

1. Clarity and Preciseness

- Good research begins with clear objectives and research questions. It's imperative to articulate what the research intends to achieve and what it seeks to explore.
- **Specific Objectives**
 - Define the goals of your research.
 - Ensure they are clear, focused, and attainable.
- **Well-Formulated Research Question**
 - Craft a question that guides your research path.
 - It should be precise and manageable in scope.

2. Relevance and Significance

- Research must contribute value to its field. It should address a gap in knowledge or solve a specific problem.
- **Identifying the Gap**
 - Understand what has already been studied.
 - Identify areas that require further exploration.
- **Impact of Research**
 - Evaluate how your research contributes to the field.
 - Consider its practical applications and theoretical implications.

3. Methodological Rigor

- The methodology is the blueprint of your research. It should be systematic, appropriate, and reproducible.
- **Appropriate Method Selection**
 - Choose methods suited to your research question and objectives.
 - Justify your methodological choices.
- **Reproducibility**
 - Ensure that others can replicate your study.
 - This enhances the reliability and validity of your results.

4. Ethical considerations

- Research must be conducted with integrity and respect for ethical norms.

- **Informed Consent**

- Obtain consent from participants, ensuring they are well-informed about the research.
- Respect their privacy and confidentiality.

- **Avoiding Bias**

- Recognize and mitigate potential biases in your study.
- Strive for objectivity and fairness.

5. Logical Structure and Coherence

- A well-structured research flows logically from one point to another.
- **Consistent Argumentation**
 - Present your arguments and findings coherently.
 - Ensure each section logically leads to the next.
- **Clear and Concise Presentation**
 - Communicate your findings effectively.
 - Use language that is accessible and clear.

6. Comprehensive Data Analysis

- The interpretation of data should be thorough and accurate.

- **Rigorous Data Analysis**

- Employ appropriate statistical or qualitative analysis methods.
- Interpret your data in a way that answers your research question.

- **Critical Evaluation**

- Assess the strengths and weaknesses of your findings.
- Discuss the implications and limitations of your research.